

ME2801: Homework 1

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1 Solving ODEs with Laplace

Taking the Laplace transform of both sides of the ODE, with zero initial conditions, yields

$$sX(s) + 7X(s) = \frac{5}{s}$$

Solving for $X(s)$

$$X(s) = \frac{5}{s(s+7)}$$

Then using MATLAB for the PFE...

```
num = 5;  
den = [1 7 0];  
% or, equivalently  
den = conv([1 0],[1 7]);  
[R,P,K] = residue(num,den)
```

```
R =  
    -0.7143  
     0.7143  
P =  
    -7  
     0  
K =  
    []
```

We have two real poles at $s = 0$ and $s = -7$. Following the notes from class, we can rewrite $X(s)$ as

$$\begin{aligned} X(s) &= \frac{-0.7143}{s+7} + \frac{0.7143}{s} \\ &= 0.7143 \left[\frac{1}{s} - \frac{1}{s+7} \right] \end{aligned}$$

Taking the inverse Laplace transform of each term yields

$$\boxed{x(t) = 0.7143[1 - e^{-7t}]\mu(t)}$$

2 Solving first-order ODE, complex roots

This is a subtle but important conceptual point about differential equations. The *solution* to a differential equation is a time-domain function that satisfies the constraints of the equation. In the context of this problem, *solving* the ODE requires finding a function $x(t)$ that when substituted into the differential equations maintains the equivalency. Solving differential equations with Laplace transforms does necessitate using the inverse transform to find the time-domain functions $x(t)$.

Taking the Laplace transform of both sides of the ODE, with zero initial conditions, yields

$$sX(s) + 7X(s) = \frac{5s}{s^2 + 2^2}$$

Solving for $X(s)$

$$X(s) = \frac{5s}{(s+7)(s^2+2^2)}$$

Then using MATLAB for the PFE...

```
% HW1, Nise 2.4
num = [5 0];
den = conv([1 7],[1 0 4]);
[R,P,K] = residue(num,den)
```

```
R =
-0.6604 + 0.0000i
```

$$\begin{aligned}
& 0.3302 - 0.0943i \\
& 0.3302 + 0.0943i \\
P = & \\
& -7.0000 + 0.0000i \\
& 0.0000 + 2.0000i \\
& 0.0000 - 2.0000i \\
K = & \\
& []
\end{aligned}$$

We have one real and two imaginary poles. Following the notes from class, we can rewrite $X(s)$ as

$$\begin{aligned}
X(s) &= \frac{2(0.33)s + 2(0.0943)(2)}{s^2 + 2^2} + \frac{-0.66}{s + 7} \\
&= \frac{0.66s + 0.377}{s^2 + 2^2} + \frac{-0.66}{s + 7} \\
&= 0.66 \frac{s}{s^2 + 2^2} + \left(\frac{0.377}{2} \right) \frac{2}{s^2 + 2^2} + \frac{-0.66}{s + 7}
\end{aligned}$$

Taking the inverse Laplace transform of each term yields

$$\boxed{x(t) = 0.66(\cos(2t)) + 0.189(\sin(2t)) - 0.66(e^{-7t})}$$